



Faculty of Engineering & Technology
Electrical & Computer Engineering
Department

ENCS3340

Project 1 Report

**Optimizing Job Shop Scheduling in a Manufacturing Plant
using Genetic Algorithm**

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Section : 2,4

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Problem Formulation:

1- Objective Function:

To minimize the makespan (total completion time).

2- Chromosome representation:

Use permutations of job sequences, where each gene represents a job and its position in the sequence represents its priority.

3- Initialize population :

Heuristic seeding → empty domain- specific heuristics to calculate initial population

⇒ pros → 1- leverage problem-specific knowledge
2- Increase the chances of finding good solution early.

⇒ Cons → 1- requires understanding of the problem.
2- May not always generalize well

⇒ Heuristic in our project that prioritizes jobs based on processing times.

4- Selection:

Implement a selection process to choose which solutions will be used to create the next generation based on their fitness which is determined by → higher fitness for lower completion time.

⇒ Two parent → the solution with highest fitness

5- Cross- over :

Define a crossover machine to combine two parents' solutions to produce offspring.

6- Mutation :

Introduce mutation to maintain genetic diversity within the population. The involve randomly swapping two jobs in the sequence.

The mutation happen with small probability equal 0.1

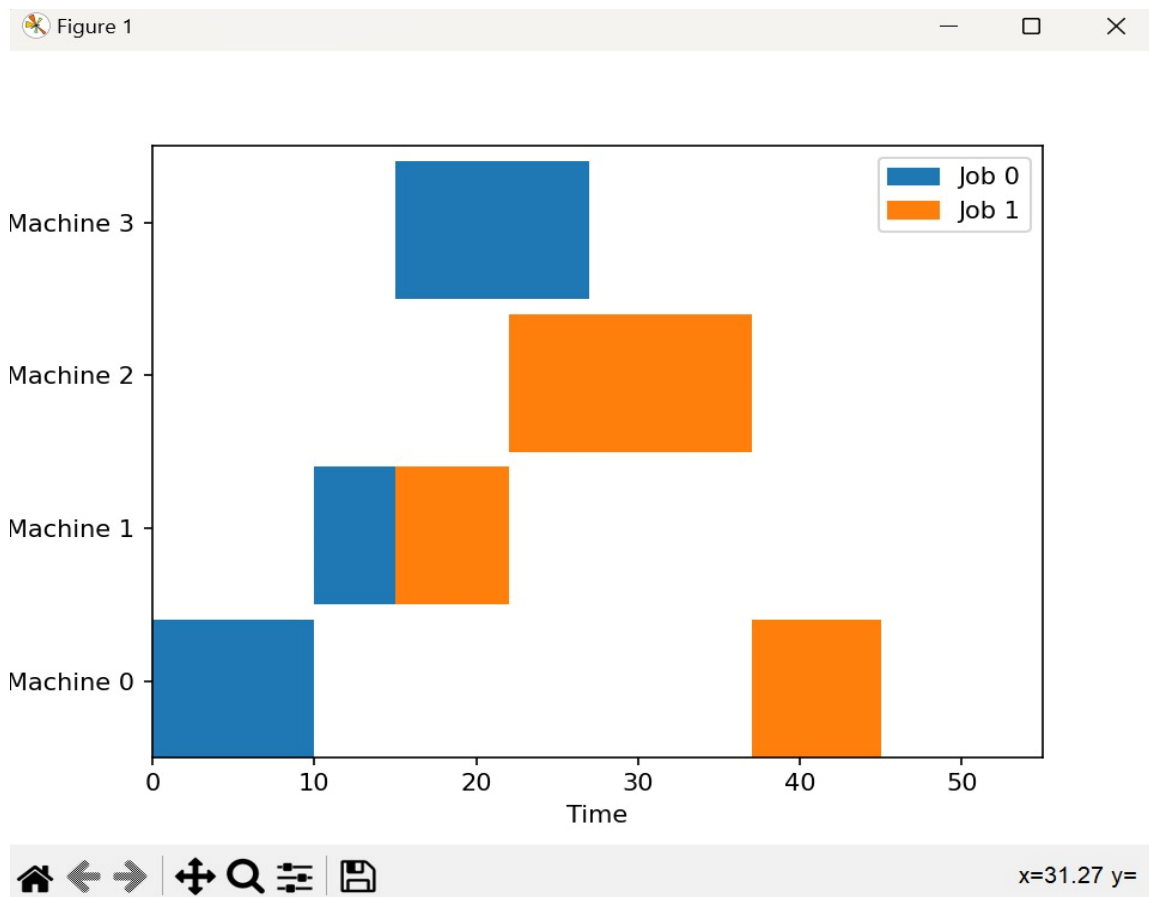
Test cases:

First test case:

The input :

```
Enter the number of jobs: 2
Enter the number of machines: 4
Enter operations for job 0 in the format [(machine, time), ...]: [(0,10),(1,5),(3,12)]
Enter operations for job 1 in the format [(machine, time), ...]: [(1,7),(2,15),(0,8)]
```

The output on Gantt Chart:

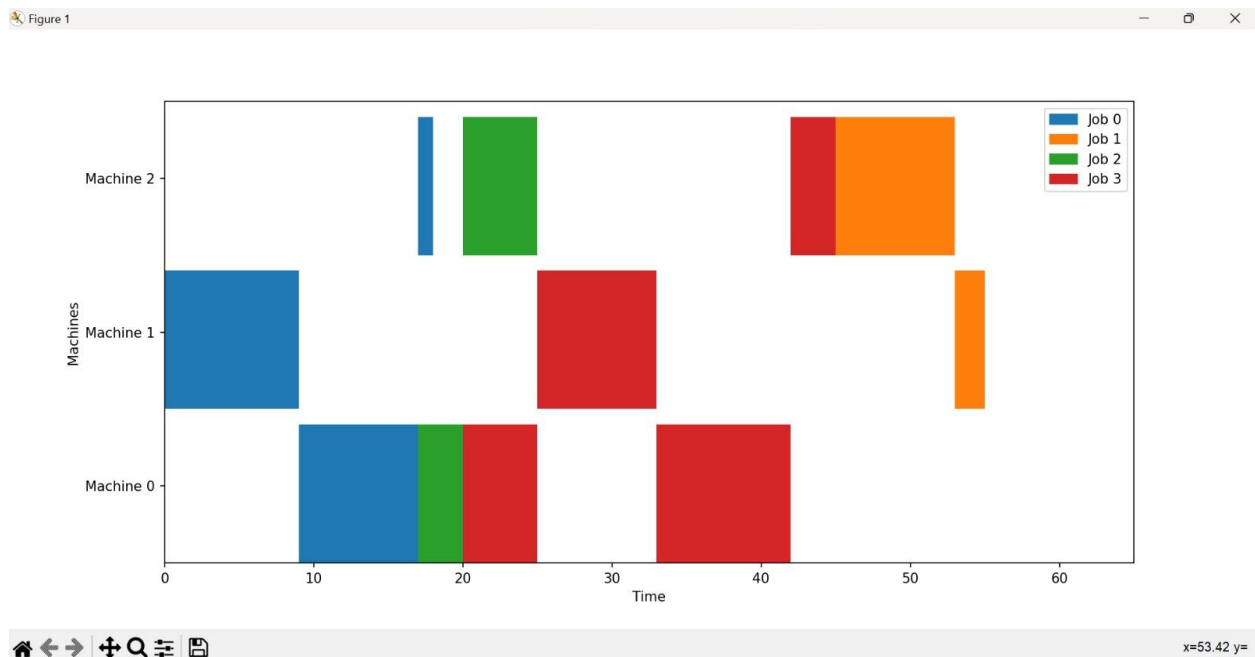


Second test:

The input:

```
Enter the number of jobs: 4
Enter the number of machines: 3
Enter operations for job 0 in the format [(machine, time), ...]: [(1,9),(0,8),(2,1)]
Enter operations for job 1 in the format [(machine, time), ...]: [(2,8),(1,2)]
Enter operations for job 2 in the format [(machine, time), ...]: [(0,3),(2,5)]
Enter operations for job 3 in the format [(machine, time), ...]: [(0,5),(1,8),(0,9),(2,3)]
```

The output on Gantt Chart:



Third test :

The input :

```
Enter the number of jobs: 2
Enter the number of machines: 2
Enter operations for job 0 in the format [(machine, time), ...]: [(0,1),(1,3)]
Enter operations for job 1 in the format [(machine, time), ...]: [(0,3),(1,4),(0,4)]
```

The output on Gantt Chart :

